

9th World Mathematics Team Championship 2018

Intermediate Level Round 1 Numerical Answers

1	2	3	4	5	6	7	8
B) 72	A) 1	C) 301	D) -1	C) 40	D) 13	E) 12	E) 85

9	10	11	12	13	14	15
B) 6	C) 4	A) 0	B) 2	D) 54	D) 81	C) 36

Intermediate Level Round 1 Detailed Solutions

1. Since $38808 = 2^3 \cdot 3^2 \cdot 7^2 \cdot 11$ the number of positive integer divisors equals

$$(3+1) \times (2+1) \times (2+1) \times (1+1) = 72$$

2. Since $2^{x+1} \cdot 4^{x+2} \cdot 8^{x+3} = 2^{x+1+2(x+2)+3(x+3)} = 2^{6x+14}$ and $16^{x+4} = 2^{4(x+4)} = 2^{4x+16}$ we have solve $6x + 14 = 4x + 16$. The solution is $x = 1$.

3. Since

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{100^2}\right) = \frac{(1 \times 3) \times (2 \times 4) \times (3 \times 5) \times (4 \times 6) \times \dots \times (98 \times 100) \times (99 \times 101)}{(100!)^2} = \frac{101}{200}$$

we have $a = 101$ and $b = 200$ giving $a + b = 301$

4. We have $\frac{1+\sqrt{12}}{1-\sqrt{12}} + \frac{\sqrt{48}}{\sqrt{121}} + \frac{2}{11} = \frac{1+2\sqrt{3}}{1-2\sqrt{3}} + \frac{4\sqrt{3}+2}{11} = -\frac{13+4\sqrt{3}}{11} + \frac{4\sqrt{3}+2}{11} = -1.$

5. We use the well-known fact that $S_{ABO} \times S_{CDO} = S_{BCO} \times S_{ADO}.$

Therefore $S_{ADO} = \frac{S_{ABO} \times S_{CDO}}{S_{BCO}} = \frac{35 \times 48}{42} = 40 \text{ cm}^2.$

6. Since $(a + bx)^3 + (a - bx)^3 = 2a^3 + 6ab^2x^2$ it follows that $2a^3 = -16$ and $6ab^2 = -108.$

The first equation implies $a = -2$ and the second one $b = 3$, so $a^2 + b^2 = 13.$

7. The shortest altitude is perpendicular to the side of 21 cm. By error and trial, having in mind $5^2 + 12^2 = 13^2$ and $12^2 + 16^2 = 20^2$, we see that the answer is 12 cm.

8. The smallest 3-digit number divisible by 7 equals $105 = 7 \times 15$ and the largest 3-digit number divisible by 7 equals $994 = 7 \times 142$. Thus the number of 3-digit numbers divisible by 7 equals $142 - 14 = 128.$

The smallest 3-digit number divisible by 21 equals $105 = 21 \times 5$ and the largest 3-digit number divisible by 21 equals $987 = 21 \times 47$. Thus the number of 3-digit numbers divisible by 21 equals $47 - 4 = 43.$

The number of 3-digit numbers divisible by 7 but not divisible by 3 equals $128 - 43 = 85.$

9. It is not possible to put a or b after ab . So, after first a in w we have at most 2 letters. Since the longest word consisting only by letter b has length 3 we conclude that the length of w is at most 6. The example of such word is $bbbaaa$.

10. Since $2x^4 - 2x^2y^2 + y^4 - 8x^2 + 20 = (x^2 - y^2)^2 + (x^2 - 4)^2 + 4$ we conclude that the least value of the given expression is at least 4. It is achievable for $x = y = 2.$

$$\begin{aligned}
11. \quad & 2018^3 - 2019^3 + 1 + 3 \times 2018 \times 2019 \\
&= 1 + 3 \times 2018 \times 2019 - (2019^3 - 2018^3) \\
&= 1 + 3 \times 2018 \times 2019 - (2019^2 + 2019 \times 2018 + 2018^2) \\
&= 1 + 2 \times 2018 \times 2019 - 2019^2 - 2018^2 \\
&= 1 - (2019 - 2018)^2 = 0.
\end{aligned}$$

12. Since $S_{APD} = S_{ABQ} = \frac{1}{2} S_{ABCD}$ we have $7+3 = 8 + x$, thus $x = 2 \text{ cm}^2$.

13. Note that $3 - 3^2 + 3^3 - 3^4 = 60$ which implies that $3^n(3 - 3^2 + 3^3 - 3^4)$ is divisible by 60 for any positive integer n . Modulo 60 the sum reduces to $3^{2017} - 3^{2018}$. Since 3^{4k+1} equals 1 modulo 60 we have that $3^{2017} - 3^{2018}$ modulo 60 equals -6 or the remainder is 54.

14. Since $A = (3-2) \times A = \frac{3^{64} - 2^{64}}{3^{60}} = 81 - \frac{2^{64}}{3^{60}}$ and

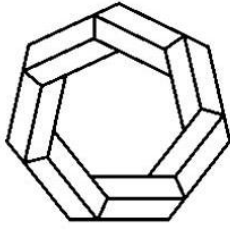
$$\frac{2^{64}}{3^{60}} < \frac{1}{2} \Leftrightarrow 2^{65} < 3^{60} \Leftrightarrow 2^{13} < 3^{12} \Leftrightarrow 8 \times 2^{10} < 9 \times 3^{10}$$

we infer that the closest integer to A is 81.

15. It follows from $1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$ that

$$10^3 + 11^3 + \dots + 20^3 = \left(\frac{20 \times 21}{2} \right)^2 - \left(\frac{9 \times 10}{2} \right)^2 = 210^2 - 45^2 = 165 \times 255 = 3^3 \times 5^2 \times 11 \times 17$$

and the sum of all prime divisors equals $3 + 5 + 11 + 17 = 36$.



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Intermediate Level Round 2 Numerical Answers

1	2	3	4	5	6	7	8
750	120	3	130	100	1009	6720	61

Intermediate Level Round 2 Detailed Solutions

1. The kilometers in which one of the tires A and B is a spare tire are 450. The kilometers in which one of the tires C and D is a spare tire are 300. So tire E has been spare tire for $1000 - 450 - 300 = 250$ kilometers and it has been in use for 750 kilometers. Such a model is possible for: $ABCE - 300$ km, $BCDE - 450$ km and $ABCD - 250$ km

2. If $\angle ACD = \gamma$ and $\angle BDC = \delta$ then

$\gamma + \delta = \angle BAO + \angle ABO = 120^\circ - \angle DAC - \angle DBC = 120^\circ - \gamma - \delta$. Thus $\gamma + \delta = 60^\circ$ and the answer is 120° .

3. After cancelling $a_n^2 = 2a_{n+1} - 1$ the sum is equal to $2a_1 - 1 = 3$.

4. The angle between the altitudes from A and B equals $180^\circ - \gamma$. Therefore $\gamma = 80^\circ$. The angle between the angular bisectors of angle A and angle B equals $90^\circ + \frac{\gamma}{2} = 90^\circ + 40^\circ = 130^\circ$. Depending on the choice of the angle between the altitudes and the angular bisectors all 140° , 150° and 40° are acceptable.

5. Since $2b$ is even we have a carryover from $c + a$. The last digit of $a + c$ is odd and there is no carryover from $2b$. So $a + c = 11, 13, 15$ or 17 and $b = 0, 1, 2, 3$ or 4 . Since $a + c = 11, 13, 15$ or 17 for $8, 6, 4$ and 2 ways we have 20 options for $a + c$; 5 options for b ; all 100.

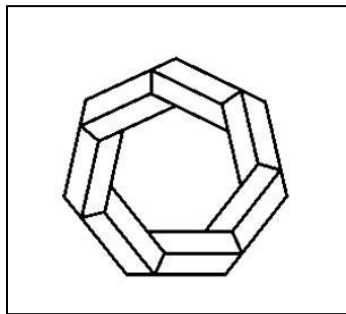
6. Let $R \neq P$ be a point on the line AB such that $BR = BP$. Triangle PRC is isosceles and PQ is middle segment in ARC . Thus, $PQ = \frac{1}{2}CR = \frac{1}{2}CP = 1009$ cm.

7. Let a_i for $i = 1, 2, 3, 4$ be the number of rings of finger i . Then $a_1 + a_2 + a_3 + a_4 = 5$ for $a_i \geq 0$ has $\binom{8}{3} = 56$ solutions. The five rings can be distributed in $5! = 120$ ways. The answer is $56 \cdot 120 = 6720$.

8. Let the number of problems from day 1 to day 11 inclusive be x ; from day 12 to day 14 inclusive be y and from day 15 to day 31 inclusive be z . The condition of the problem gives $x + y = 5 \cdot 14 = 70$ and $y + z = 3 \cdot 20 = 60$. Thus

$$n = x + y + z = 130 - y.$$

Since $0 \leq y \leq 60$ the number of different values of n is 61.



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Intermediate Relay Detailed Solutions

R1-A Sum of 14 is achieved by $6 + 6 + 2$; $6 + 4 + 4$; $5 + 5 + 4$ and $6 + 5 + 3$.

For each of the first three triples the probability $3 \times \frac{1}{6^3}$ and for the last one it is $6 \times \frac{1}{6^3}$

. The probability is $3 \times 3 \times \frac{1}{6^3} + 6 \times \frac{1}{6^3} = \frac{15}{6^3} = \frac{5}{72}$, giving $p = 5$, $q=72$ with

$$p + q = 77.$$

Answer: 77

$$T = 77$$

R1-B Let $\angle ACB = \gamma = T$, $\angle ABC = \beta$ and $\angle BAC = \alpha$. Since PM and QM are medians in the right triangles ABP and ABQ we have that $BM = PM = QM = AM$. Therefore $\angle BMP = 180^\circ - 2\beta$ and $\angle AMQ = 180^\circ - 2\alpha$ implying

$$\begin{aligned} \angle PMQ &= 180^\circ - (180^\circ - 2\beta) - (180^\circ - 2\alpha) = -180^\circ + 2(\alpha + \beta) = -180^\circ + 2(180^\circ - \gamma) \\ &= 180^\circ - 2\gamma = 180^\circ - 2(77^\circ) = 26^\circ. \end{aligned}$$

Answer: 26

R2-A Since $b + c > a$ we have that $|a - b - c| = b + c - a$ and therefore

$$||a - b - c| - c| + |a - b - c| = |b - a| + b + c - a$$

If $a > b$ the above expression equals c and if $a < b$ it is $2(b - a) + c$. The greatest value equals 2017 and it is achieved for $a = 2012$, $b = 2014$ and $c = 2013$

Answer: 2017

$$T = 2017$$

R2-B The distance travelled by B is two times the distance travelled by A . The two points meet for the first time at point X such that the arc BX is two times the arc AX . After that they meet at B then at Y then at X and so on.

Since $T = 2017 = 3 \times 672 + 1$ they meet for the last time at X .

Answer: 60

R3-A

$$\begin{aligned} & \sqrt{3 - \sqrt{4 + \sqrt{12}}} + \sqrt{3 + \sqrt{4 - \sqrt{12}}} \\ &= \sqrt{3 - \sqrt{(1 + \sqrt{3})^2}} + \sqrt{3 + \sqrt{(1 - \sqrt{3})^2}} \\ &= \sqrt{2 - \sqrt{3}} + \sqrt{\sqrt{3} + 2} = \sqrt{\frac{(\sqrt{3} - 1)^2}{2}} + \sqrt{\frac{(\sqrt{3} + 1)^2}{2}} = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6} \end{aligned}$$

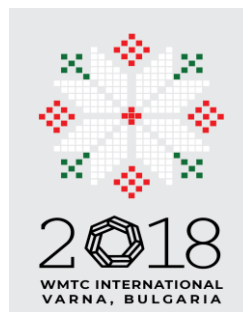
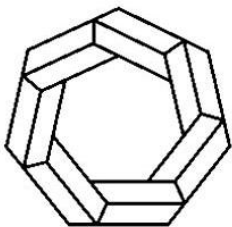
giving $5x^2 = 30$.

Answer: 30

$$T = 30$$

R3-B The induced graph has no triangles. The maximum number of edges in a graph of $T = 30$ vertices without triangles equals $15^2 = 225$. The maximum number of presents is $2 \times 225 = 450$.

Answer: 450



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Intermediate Team Detailed Solutions

1. The set $\{1, 2, 3, 5, 8\}$ has the desired property. Assume there exists a set $A = \{a_1, a_2, a_3, a_4, a_5\}$, with $a_5 < 8$ the largest, satisfying the property. If $1, 2 \in A$ then among the remaining three numbers there are no consecutive numbers. The only such case is $3, 5, 7 \in A$ and then $1+7=3+5$. So at least one of 1 and 2 is not an element of A . By the similar arguments at least one of 7 and 6 is not an element of A . Then $3, 4, 5 \in A$ and easy verification of all 4 cases leads to a contradiction.

Answer: 8

2. Note that $2018 = 2 \times 1009$ and 1009 is a prime number. Also for $n = \underbrace{1999 \dots 99}_{112}$ we have $S(n) = 1009$ and $S(n+1) = 2$. The sum of the digits of $n-1 = \underbrace{1999 \dots 98}_{112}$ is 1008, so $n-1$ is not a solution. Assume $S(m)S(m+1) = 2018$ for $m < n-1$. The sum of the digits of any positive integer, less than $n-1$ is less than 1009. So both $S(m)$ and $S(m+1)$ are less than 1009, a contradiction.

Answer: $n = \underbrace{1999 \dots 99}_{112}$.

3. Denote $S_{ABD} = x$, $S_{BDX} = S_{ADY} = y$ and $S_{DXY} = z$. We have

$$x + y = S_{ABX} = \frac{BX}{BC} \times S_{ABC} = \frac{4}{6} \times 18 = 12 \text{ and } y + z = S_{AXY} = S_{ABC} - 12 - 2 = 4.$$

We use that $x = 12 - y$, $z = 4 - y$ and $y^2 = x \times z$. Thus $y^2 = (12 - y)(4 - y)$ which implies $y = 3$. Thus $x = 9 \text{ cm}^2$.

Answer: 9

4. Denote the number of words of length n with the given property by T_n .

Partition the words from T_n in two sets: words with last letter b (denote this set by T_n^b) and words with last letter a (denote this set by T_n^a). It is clear that $T_n^b = T_{n-1}$ and $T_n^a = T_{n-2}$. So $T_n = T_{n-1} + T_{n-2}$. From $T_1 = 2$ and $T_2 = 3$ we obtain: $T_3 = 5$, $T_4 = 8$, $T_5 = 13$, $T_6 = 21$ and $T_7 = 34$.

Answer: 34

5. Assume all cells have odd number of adjacent cells. By considering cells (1,1) and (2,2) it is easy to see that there are even number of colored cells among (3,2) and (2,3). The same arguments for (4,4) and (5,5) imply that there are even number of colored cells among (3,4) and (4,3). Now the cell (3,3) has even number of colored adjacent cells. The example of exactly one cell with even number of colored adjacent is given by the table.

X	X		X	X
		X		
	X	X	X	
		X		
X	X		X	X

Answer: 1

6. The first minute each elephant drank 4 liters; the second minute each elephant drank 5 liters and the third minute each one drunk 8 liters. Each of the last 5 has $4 + 5 + 8 = 17$ liters.

Answer: 17

7. For 1 min the big clock arrow travels for halve degree. For 1 min the small clock arrow travels for 6 degrees. So the angle between big clock arrow and 12 o'clock direction equals $210^\circ + 19^\circ = 229^\circ$. The angle between small clock arrow and 12 o'clock direction equals $38 \times 6 = 228^\circ$. Then the angle between two arrows equals 1° .

Answer: 1

8. It follows from isosceles triangle AMN and $\angle BAC = 20^\circ$ that $\angle AMN = 20^\circ$. Then $\angle CNM = 40^\circ$ and $NM = MC$ implies $\angle NCM = 40^\circ$. From triangle ANM we obtain $\angle CMN = 100^\circ$. Now $\angle BMC = 60^\circ$ and triangle MBC is equilateral. Therefore $MB = MC = MN$ and it follows from isosceles triangle MBN that $\angle BNM = 10^\circ$.

Answer: 10

9. For $x \in \left[\frac{1}{2}; \frac{2}{3}\right]$ the equation is equivalent to $(2x-1)-(3x-2)=a-x$ or $a=1$.

Answer: 1

10. $T = 1$. Since $\overline{c1b}$ is divisible by 5 we have that $b = 0$ or $b = 5$. But $b \neq 0$ thus $b = 5$. Since $\overline{15c}$ is divisible by 4 we have that $\overline{5c}$ is divisible by 4, so $c = 2$ or $c = 6$.

Since $\overline{521}$ is not divisible by 3 we have $c = 6$.

Answer: 6

11. $T = 1$. Write the equation $x + y = xy - 7$ in the form $(x-1)(y-1) = 8$.

Since x is positive integer and $x-1$ is a divisor of 8 it can take 4 values: 2, 3, 5 and 9. The corresponding values of y are: 9, 5, 3 and 2.

Answer: 4

12. $T = 12$. The number of games equals 21. Each game contributes to the total number of points by 3 or 2. Therefore the total number of points can be any number between 42 and 63. Since it is divisible by 24 it equals 48. Then the number of wins is $48 - 42 = 6$.

Answer: 6

13. $T = 9$. Note that 111 and 111111 are both divisible by 3 but not by 6. It is easy to see that 111111111 is divisible by 9 but not by 27. Using the above, $T = 9$ and

$$3 \times 33 \times 333 \times \dots \times \underbrace{33\dots33}_9 = 3^9 \times 1 \times 11 \times 111 \times \dots \times \underbrace{11\dots11}_9$$

we conclude that $k = 13$.

Answer: 13

14. $S = 17, T = 10$. Denote the speed of the student by x and the speed of the tram by y . If the distance between two trams is D then $\frac{D}{x+y} = T = 10$ and $\frac{D}{y-x} = S = 17$

. Therefore $\frac{x+y}{y-x} = \frac{17}{10}$ which is equivalent to $27x = 7y$ or $\frac{x}{y} = \frac{7}{27}$.

Answer: $\frac{7}{27}$